GitHub Actions & The GitHub ↔ AWS Cloud Bridge (Visual Architecture Guide)

How Code Moves from Local Git Push & Pull Requests ➔ Automated GitHub Actions Robots ➔ Live AWS Production Cloud (us-east-2)

Repo: cosmosanalytics/AWS

AWS Project: 966273137102 (us-east-2)

1. HOW CODE ENTERS GITHUB: GIT PUSH VS. PULL REQUEST

GIT WORKFLOW

git push (Direct Upload)

Uploads commits straight from your laptop to a GitHub branch (**`git push origin main`**).

- **Speed:** Instant — no waiting for human review.
- **When used:** Sole owner pushing directly to **`main`** or uploading to a sandbox branch.

S&OP Analogy: Directly updating the live plan as the system owner.

Pull Request / PR (Review Gate)

A formal request to merge a **`feature-branch`** into **`main`** after automated checks pass.

- **Safety:** Runs line-by-line AI diff review + CI tests *before* touching **`main`**.
- **Human Gate:** Pauses for explicit approval.

S&OP Analogy: Promoting a *Private Sandbox Scenario* to the *Committed Plan*.

2. WHAT IS GITHUB ACTIONS? (4 AUTOMATED ROBOT INSPECTORS ON UBUNTU-LATEST)

100% FREE ON PUBLIC REPOS

JOB 01 • AST AUDIT

AI Code Review

Boots a Linux VM & verifies all 3 Tiers and all 9 Bedrock models (Claude 4.6, DeepSeek, etc.) exist in **`index.html`**.

JOB 02 • DEVSECOPS

Zero-Leak Shield

Scans code to guarantee zero leaked AWS keys (**`AKIA...`**) and verifies the **`X-Cosmos-Key`** passcode gate.

JOB 03 • CI MATRIX

48/48 Unit Tests

Executes 48 automated assertions checking every canvas node, autonomy policy (**`p1-p8`**), and rollback rule.

JOB 04 • PRE-FLIGHT

AWS Gate Ping

Pings your live AWS **`us-east-2`** Lambda URL to confirm it is online and enforcing **`401`** security for strangers.

Why Is It Free? Public repos (`cosmosanalytics/AWS`) get **UNLIMITED free GitHub Actions minutes!** (Private repos: 2,000 free mins/mo, then $0.008/min)

3. HOW GITHUB & AWS CONNECT: EXACT DIRECTION OF INFORMATION FLOW (GITHUB ➔ AWS VS. AWS ➔ GITHUB)

SHA-256 OWNER PASSCODE GATE

GitHub Cloud — cosmosanalytics/AWS

1. Git Repository (main & gh-pages)
Stores **`index.html`**, **`pages.yml`** workflow, and commit history (**`#a1202bb`**).

2. GitHub Actions Runner VM
Runs Jobs 1–4 on every push & exposes run status via **`api.github.com`**.

3. GitHub Pages Web App (CDN)
Serves the interactive 3-Tier CI/CD Canvas at **`cosmosanalytics.github.io/AWS/`**.

① FORWARD TRIGGER (PUSH): GitHub Pages / Actions ➔ AWS Lambda

INFO FLOW: GITHUB ➔ AWS

Browser UI (unlocked with **`cosmos2026`**) or GitHub Actions Job 4 sends an encrypted **HTTPS POST** request with **`X-Cosmos-Key`**, selected model (**`Claude Sonnet 4.6`**), and audit prompt to your AWS Lambda Function URL in **`us-east-2`**.

② LIVE TELEMETRY FETCH (PULL): AWS Lambda Queries GitHub REST API

INFO FLOW: GITHUB ➔ AWS

AWS Lambda calls **`GET https://api.github.com/repos/cosmosanalytics/AWS/commits`** & **`/actions/runs`**. *(Even though AWS initiates the call, the data—live Commit SHA **`#a1202bb`** & Run **`#39148195073`**—flows **from GitHub into AWS**!)*

③ REVERSE RECEIPT & STATUS FLOW: AWS us-east-2 ➔ GitHub App & Repo

INFO FLOW: AWS ➔ GITHUB

AWS returns the live **Claude 4.6 Audit Text**, **Step Functions Execution ARN** (**`exec-...`**), and **S3 Artifact URI** back to the GitHub Pages Web UI (and can post a green ✅ **`Deployed`** commit status / PR comment back to the GitHub repo).

AWS Cloud (us-east-2) — Proj: 966273137102

1. AWS Lambda Orchestrator
`cosmos-cicd-ai-orchestrator` verifies SHA-256 passcode before any cloud spend.

2. Amazon Bedrock (9 Live FMs)
`Claude Sonnet 4.6`, **`Opus 4.6`**, **`Haiku 4.5`**, DeepSeek V3.2, Mistral Large 3, GPT-OSS 120B.

3. Step Functions & S3 Store
Runs **`cosmos-cicd-ai-pipeline-sfn`** & saves JSON receipt to **`s3://cosmos-ai-factory-cicd...`**

For Non-Owners (Public LinkedIn Visitors) — SIMULATED MODE ($0.00 Cost)

100% SAFE & FREE

Simulated GitHub: Plays back cached commit **`#2588884`** & Actions Run **`#39148195073`** locally without hitting GitHub API rate limits.

Simulated AWS: Plays back authentic pre-recorded receipts for all 9 Bedrock models locally in the browser (**`$0.00`** AWS bill).

Full UI Interactivity: Visitors can test all 3 Hierarchy Levels, Human Approval Gates, Canary Sliders, and Rollback Ledgers.

Server-Side Firewall: Unauthorized calls to the Lambda URL are blocked at the door with **`HTTP 401 Unauthorized`**.

For You (Owner Passcode Unlocked) — 100% REAL GITHUB + REAL AWS MODE

LIVE CLOUD EXECUTION

Real GitHub REST API: Pulls live commits, branch heads (**`main/gh-pages`**), and real GitHub Actions runs in real time.

Real Amazon Bedrock (us-east-2): Invokes live **`Claude Sonnet 4.6`** (or any of the 9 models) with the live GitHub commit context.

Real AWS Step Functions: Starts a new state machine execution (**`exec-...`**) stamped with the live GitHub SHA.

Real Amazon S3 Persistence: Writes the permanent JSON deployment & audit receipt to your S3 bucket in **`us-east-2`**.